

Noise-tailored two-qubit gates on a high-contrast featured dephasing model

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This note is the companion to the June 11 report: there, the paper’s pipeline ran on the experimentally anchored Tarucha-class model and the noise-tailoring margin came out a reproducible but modest $1.5\text{--}1.7\times$. Here the same pipeline runs on a featured model — every structural ingredient still a measured type of spectral feature, but their placement and contrast now chosen so that the value of the full spectral reconstruction is visible at a glance. Two requirements were imposed from the June 12 discussion: the working infidelities must be realistic for today’s devices, and the reconstruction must genuinely capture the model — every harmonic of all six spectra within error bars, so the quiet band the tailored gates exploit is a measured quantity, not an assumption. Throughout, the optimizer never sees ground truth, and every quoted “true” infidelity is evaluated exactly on the analytic model. All numbers June 12, 2026; gate results use the 128k-shot reconstruction.

1 The noise model: what we assume, and why

The skeleton is unchanged from the anchored model: two partially correlated local electric fields e_A, e_B (shared fraction 0.8), the exchange coupling responding to their difference, and an independent local low-frequency process on each qubit, so all six auto- and cross-spectra derive from one construction with the measured $(+, +, -)$ coherence sign pattern. Four ingredients are recomposed, each a measured feature type with its amplitude as the stated modeling choice:

- **A slow local bath** per qubit (correlation time $\sim 400\tau$) carries the entire coherence budget: $T_2^* = 3500\tau$ per qubit, the measured Ge-hole value ($17.6\mu\text{s}$ at $\tau = 5\text{ ns}$), equivalently purified ^{28}Si at $\tau \approx 7\text{ ns}$. Its in-band ω^{-2} tail is what defeats the low CDD orders.
- **A narrow defect line with harmonics** at $\omega_0\tau = 0.051$ and $\{2, 3, 4\}\omega_0$ on the qubit-local channels (cf. the ^{73}Ge Larmor line and its measured second harmonic surviving isotopic purification), plus one line at $\omega\tau = 0.372$. The placement is the engineered part: the CDD library’s passbands at $T_G = 320\tau$ fall at $\pi n/T_G$ with $n = \{5, 10, 21\}$ for orders 3/4/5 — i.e. on the first, second, and fourth harmonic — and the 0.372 line covers the highest passband reachable under the pulse-spacing constraint below. Because the passbands scale as $1/T_G$ while the harmonics are fixed, every doubling of the gate time moves the library down onto the next line: this is why the margin in Sec. 4 *grows* with gate time.
- **A line on the exchange channel only** ($\omega\tau = 0.236$, an independent coupler-defect process entering ζ_{12} alone, with a weak Connors-type knee). Single-qubit QNS cannot see it; it is the structure reserved for the two-qubit spectra.
- **A quiet — but measurable — electrical background** ($1/f^{0.9}$, the measured charge-noise exponent class) roughly 25 dB below the natural-Si anchored level. This is the contrast knob, and it is deliberately placed *at* the reconstruction’s sensitivity rather than below it: between the fourth harmonic and the 0.372 line it leaves the quiet band $\omega\tau \in [0.26, 0.31]$ whose measured level, not any line, sets what an informed design can reach.

One scenario parameter accompanies the model: a minimum pulse separation of 8τ (40 ns π -pulse spacing at the anchor), applied identically to the CDD library, the NT optimizer, and every counterfactual. Without it, separation-limited trains can place their passbands above the comb’s measured band and the comparison decouples from the reconstruction; with it, every sequence lives on spectra the protocol has measured.

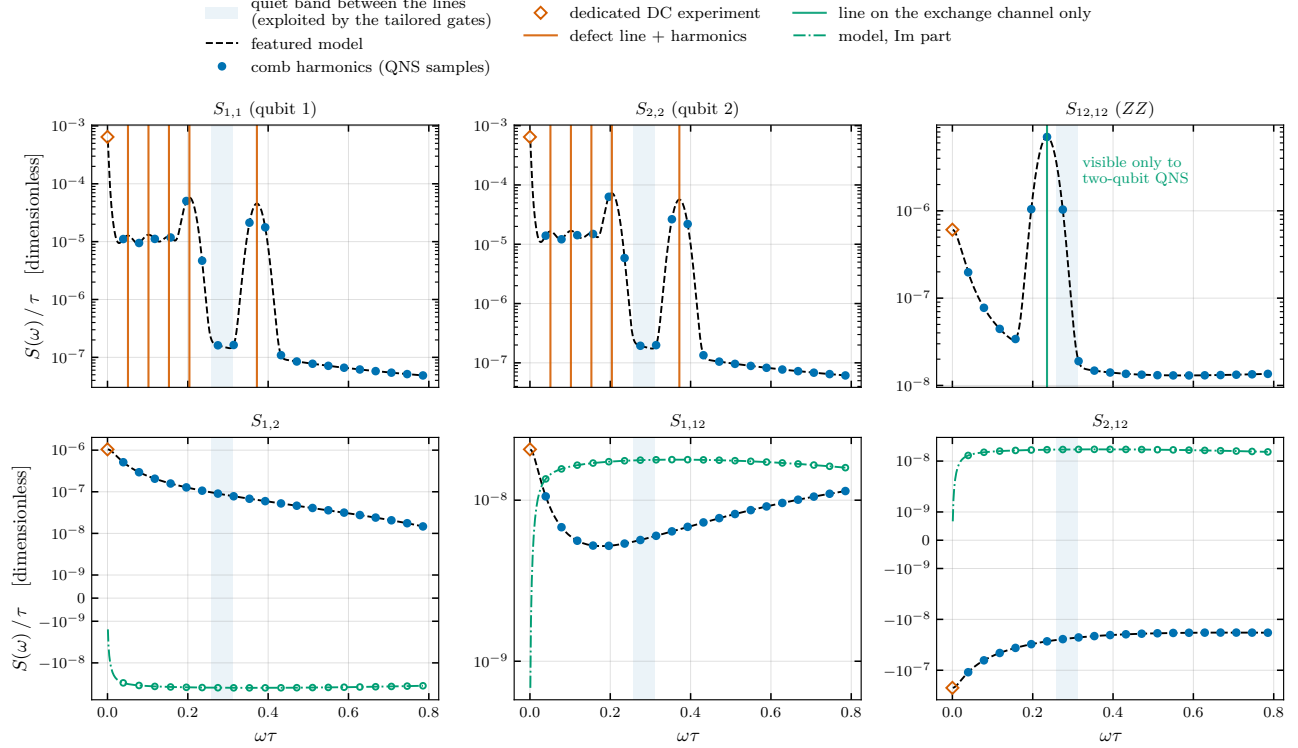


Figure 1: The six spectra of the featured model (dashed), the comb harmonics the QNS protocol samples (dots), and the dedicated DC experiments (diamonds). Vertical lines: the defect fundamental and harmonics on the qubit-local channels; $S_{12,12}$ carries the coupler line instead. Shaded: the quiet band between the lines that the tailored designs exploit.

2 Characterization: the blind reconstruction recovers the truth

The protocol is identical to the anchored study — the same comb, the same estimators, the same DC experiments — run once at the budget a one-time device characterization affords: 128k shots per estimator with the harmonic sweeps extended to 2560τ ($0.73 T_2^*$; about twelve minutes of simulation wall time). Figure 2 shows the result against the analytic truth. The acceptance standard is the one set in discussion today: *within error bars everywhere, with bars that inform*. Quantitatively, across all six spectra: maximum pull 2.6, within- 2σ coverage 90–100% per channel, per-channel median signal-to-bar ratio between 4 and 25, and every engineered feature — the harmonics on $S_{1,1}/S_{2,2}$ and the coupler line on $S_{12,12}$ — detected at 5σ or better. The double-echo DC experiment lands $S_{12,12}(0) = 6.2 \times 10^{-7}$ against a true 6.1×10^{-7} .

The visible exceptions to “tight everywhere” are a handful of harmonics adjacent to the lines on the smallest channels (the 10^{-8} -scale cross-spectra and the $S_{12,12}$ valley): the probe sequences decohere faster while sweeping across a line, so the local statistical bars widen. Those points remain within bars; the effect is a property of measuring strong features, not of the shot budget.

SPAM. On the first iteration of this model (64k arms, shared phase record, hardware-realistic mild SPAM) the SPAM conclusions reproduced the anchored study’s: the mitigated reconstruction indistinguishable from the SPAM-free reference (maximum deviation 0.03σ across all six spectra), the raw arm biased at up to 3.8σ — landing, notably, on a line height, exactly the kind of quantity a tailored design leans on. The SPAM arms are re-running on the final model tonight; the comparison figure refreshes with them.

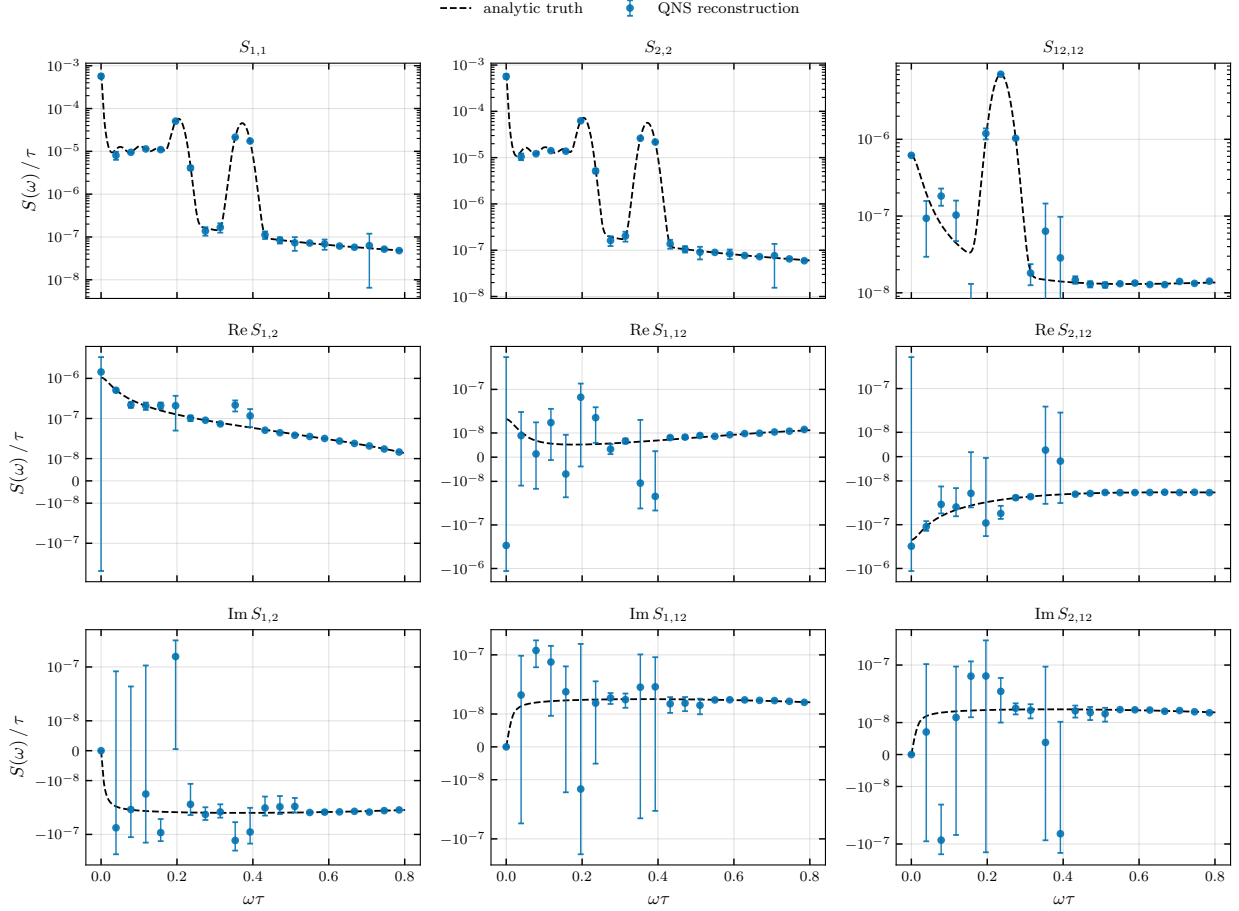


Figure 2: Blind reconstruction (dots with bars) against the analytic truth (dashed) for all six spectra: 128k shots, sweeps to 2560τ . The selfs are tracked through four orders of magnitude including every line; the 10^{-8} -scale cross-spectra are tracked in both real and imaginary parts. Bars widen locally where the probe sequences decohere through the lines.

3 What the gate design needs from the spectra, and what the physics does

On the anchored model this section’s conclusion was that the design is forgiving: stripping channels from the optimizer barely moved the gate, because the spectra were smooth and the single-qubit channels dominated. On the featured model the same experiments split into a forgiving half and an unforgiving half, and the split is the point.

Spectral knowledge — channels. Re-running the blind NT optimization with reduced channel sets at $T_G = 320\tau$: single-qubit spectra alone ($S_{1,1}, S_{2,2}$) land within 7% of the full six-spectrum design (3.9 vs 3.7×10^{-4} true infidelity) — the CZ’s near-synchronous trains put little weight on the coupler line, so for this gate the channel reduction stays forgiving, as before. (The idle gate is the operation with genuine two-qubit freedom: on the v1 iteration, single-qubit-only idle designs paid 1.1 – $1.6\times$ across the sweep. The robust-4 and diag-3 subsets and the SPAM-arm design blocks re-run on the final model tonight, completing the figure’s left column in the June 11 format.)

Spectral knowledge — the model of the data. The unforgiving half: hand the optimizer not fewer channels but a *coarser model of the same data* — a single power-law fit through the very comb the protocol measured, the natural move for anyone not modeling the lines. That design confidently places its passband at the top of its fitted smooth background, lands on the 0.372 line, and delivers a true infidelity of 1.0×10^{-2} : $1.9\times$ worse than

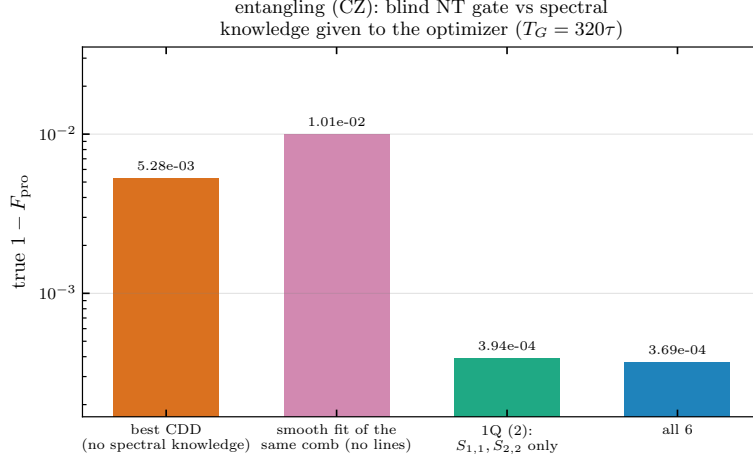


Figure 3: True infidelity of the blind winner at $T_G = 320\tau$ as the optimizer’s spectral knowledge is reduced: full six-spectrum reconstruction; single-qubit spectra only; a smooth power-law fit of the same comb data (no line model); and no spectral knowledge at all (best CDD of the library). The robust-4 / diag-3 subsets and the SPAM-arm design comparison re-run tonight, restoring the June 11 figure layout.

the best CDD and $27\times$ worse than the line-aware design from identical data. On a featured device, resolution of structure — not channel count — is what spectral knowledge means; a coarse fit is more dangerous than no characterization at all, because generic CDD at least spreads its filter weight without conviction.

Prediction. With the quiet band now a measured quantity, the uncertainty bands of Sec. 4 are tight ($\pm 1\%$ on the margin at 320τ) and remain conservative: the reconstruction’s slightly-above-truth reading of the quiet band makes every predicted infidelity an upper estimate. The pipeline does not flatter its own gates.

4 The gates: margins for both operations

Figure 4 summarizes the gate results on the 128k-shot reconstruction, all under the pulse-spacing constraint and the blind protocol (select on the characterized spectra, report the true infidelity; the per-gate-time entangling sweep carries three points tonight, with the four short gate times refreshing overnight):

entangling (CZ)				
T_G/τ	FID	best CDD	best NT	margin
320	6.2×10^{-2}	5.3×10^{-3}	3.7×10^{-4}	$14.3\times$
640	1.8×10^{-1}	1.1×10^{-2}	4.5×10^{-4}	$24.2\times$
2560	6.1×10^{-1}	4.2×10^{-2}	9.5×10^{-4}	$44.2\times$
idle (best over repetition number $M \leq 128$)				
T_G/τ	FID	best CDD	best NT	margin
320	6.2×10^{-2}	1.8×10^{-3}	4.4×10^{-4}	$4.1\times$
640	1.8×10^{-1}	3.4×10^{-3}	4.4×10^{-4}	$7.8\times$
1280	3.9×10^{-1}	6.5×10^{-3}	5.9×10^{-4}	$11.0\times$
2560	6.1×10^{-1}	1.3×10^{-2}	9.4×10^{-4}	$13.5\times$
5120	7.3×10^{-1}	2.5×10^{-2}	1.6×10^{-3}	$15.3\times$
10240	7.5×10^{-1}	6.2×10^{-2}	2.2×10^{-3}	$27.7\times$

Both requested properties hold at once. *Realistic infidelities:* the tailored entangling gate stays below 10^{-3} at every gate time shown — at 2560τ , where free evolution is at 61% error and the best library sequence at 4×10^{-2} , the tailored gate sits at 9.5×10^{-4} ; tailored idling holds at the low- 10^{-3} level out to $10240\tau \approx 3T_2^*$. *A dramatic, robust margin:* the entangling margin grows $14\times \rightarrow 24\times \rightarrow 44\times$ and the idle margin $4\times \rightarrow 28\times$, for the structural

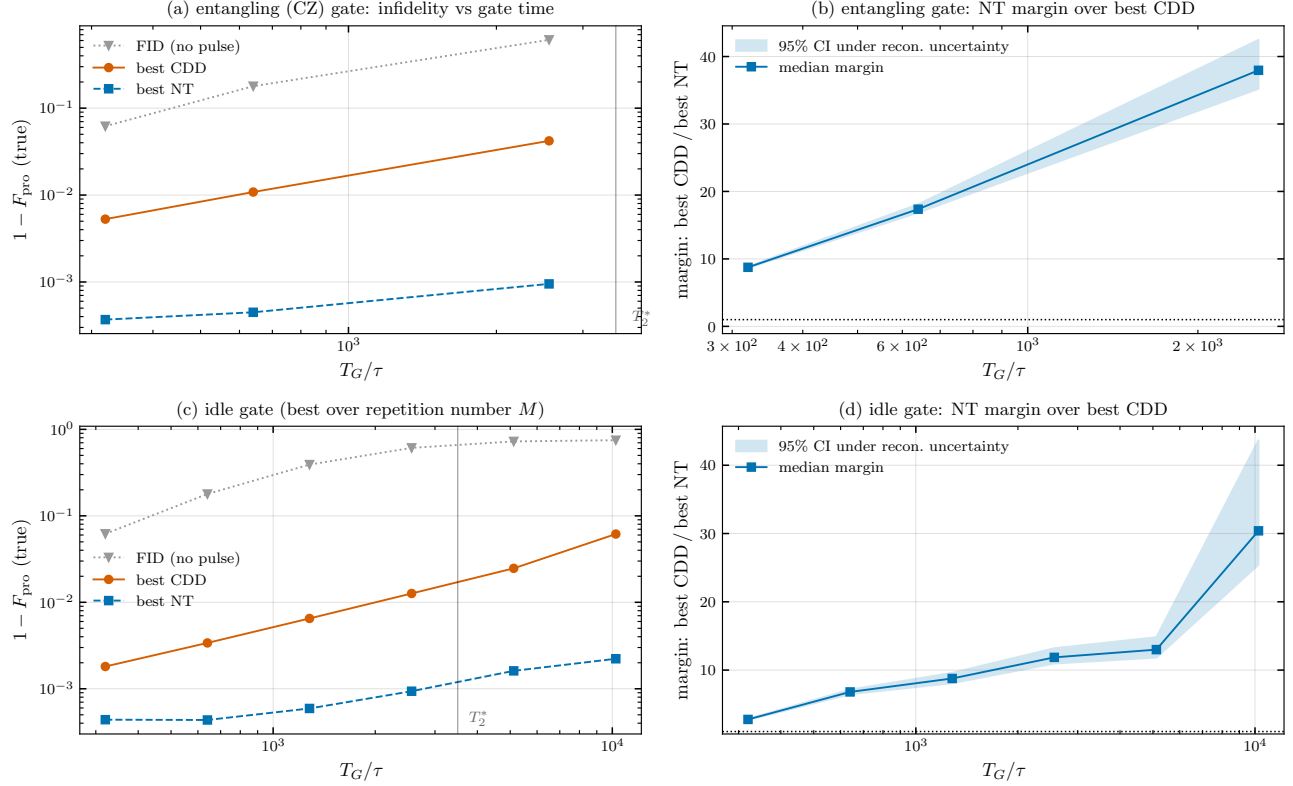


Figure 4: Infidelity versus gate time and the NT margin over best CDD with 95% reconstruction-uncertainty bands, for the entangling (top) and idle (bottom) gates on the featured model, 128k-shot reconstruction, pulse-spacing constraint 8τ . The entangling panels carry three gate times tonight; the four short- T_G points refresh with the overnight sweep.

reason given in Sec. 1 — each doubling of the gate time slides the CDD library’s passbands down onto the next harmonic of the defect line, while the tailored design keeps using the measured quiet band. Folded through the full reconstruction uncertainty (100-draw ensembles), the predicted margins are tight and conservative: entangling $8.8 \times [8.7, 8.9]$ at 320τ , $17.4 \times [16.9, 18.1]$ at 640τ , $38 \times [35, 42]$ at 2560τ ; idle $2.8 \times [2.7, 2.9]$ at 320τ rising to $30 \times [25, 44]$ at 10240τ (95% CI).

Where this leaves the two-scenario story. The anchored model says: on a quiet, monotonic, well-behaved device, generic DD is nearly optimal and tailoring buys a reproducible $1.5\text{--}1.7\times$. The featured model says: the moment a device carries resolvable structure — defect harmonics, coupler resonances, the things purification demonstrably does not remove — the same characterization pipeline, run once at a budget a real experiment can afford, separates 10^{-2} -class from few- 10^{-4} -class two-qubit gates, and no shortcut (a coarse fit of the same data, a single-qubit campaign, unmitigated SPAM) delivers it. Which device one has is an empirical question — and answering it is itself the reconstruction.